AI-BASED MICROSCOPY IMAGE ANALYSIS FOR PREDICTING URINARY TRACT AND RENAL DISEASES

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Final Group Report

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DECLARATION

ABSTRACT

The presented research proposes a new AI-based system of automated detection and diagnosis of UTI and other renal disorders using urine microscopy. Manual microscopic examination of urine is laborious and dependent on laboratory specialists; hence, results can be inconsistent and require substantial time to be obtained. In order to solve the problem, a framework combining deep learning and clinical reasoning algorithms is used in the process of identifying the most important urine components white blood cells (WBCs), red blood cells (RBCs), microorganisms (bacteria, yeast), crystals, and casts. The system is built on lightweight models that allow accurate real-time prediction of urine microscopic analysis parameters. As was confirmed by experiments, the algorithm is capable of detecting WBCs with mAP@0.5 score of 0.96, yeast with the score higher than 0.94, and classifying crystals with an accuracy of 99.41%. RBCs are separated into morphologic types, and causes of hematuria are detected with recall of 81.40%. Finally, the algorithm is capable of identifying urinary casts with the same mAP@0.5 value of 0.80.

Keywords: Urine Microscopy, AI in Diagnostics, Urinary Tract Infections (UTIs), Hematuria, Glomerulonephritis, Kidney Stones, Red Blood Cells (RBCs), White Blood Cells (WBCs), Urinary Casts, Urinary Crystals, Bacteria Classification (Cocci, Bacilli, Spirochetes), Deep Learning, Patient Metadata Integration, Image-Based Diagnostics, Explainable AI, Medical Image Analysis, Multimodal AI System, Renal Disease Detection, Clinical Decision Support, Digital Urinalysis

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
AWS	Amazon Web Services
CNN	Convolutional Neural Network
WBC	White Blood Cells
RBC	Red Blood Cells
UTI	Urinary Tract Infection
YOLO	You Only Look Once
UNet	U-Net Architecture
mAP	Mean Average Precision
IoU	Intersection over Union
F1-score	Harmonic Mean of Precision and Recall
ML	Machine Learning
DL	Deep Learning
U-Net++	Nested U-Net Architecture
B0 / B2 / B3	EfficientNet Model Variants
XAI	Explainable Artificial Intelligence
AUC	Area Under the Curve
ROC	Receiver Operating Characteristic
FastAPI	Fast API Framework
RBC Morphology	Red Blood Cell Shape Analysis

1. INTRODUCTION

1.1 Background Literature

Urine microscopy plays a critical role in diagnosing urinary tract and renal diseases by analyzing sediment components such as red blood cells (RBCs), white blood cells (WBCs), bacteria, crystals, and casts. These components act as important biomarkers for conditions including urinary tract infections (UTIs), kidney stones, hematuria, and renal disorders. However, traditional manual urinalysis is time-consuming, subjective, and highly dependent on laboratory expertise, leading to variability and inefficiency, particularly in low-resource settings.

Recent advancements in Artificial Intelligence (AI) and Machine Learning (ML), especially deep learning techniques, have enabled automated analysis of medical images with high accuracy. However, most existing studies focus on individual components rather than providing a comprehensive diagnostic framework. Additionally, limited integration of patient metadata reduces clinical interpretability. To address these gaps, this research proposes a multimodal AI-based system combining image analysis with clinical data for improved diagnostic accuracy.

1.1.1 UTI Detection using WBCs, Bacteria, and Yeast

Urinary Tract Infections (UTIs) are among the most frequently occurring infections in the urinary system, commonly caused by bacterial pathogens such as *Escherichia coli* [20]. In urine microscopy, UTIs are primarily identified through the presence of white blood cells (WBCs), bacteria, and in some cases, yeast cells. Elevated levels of WBCs (pyuria) indicate an immune response and inflammation within the urinary tract, making them a strong biomarker for infection. In more severe conditions such as pyelonephritis, WBC casts may also appear, suggesting upper urinary tract involvement [22], [23].

The detection of bacteria in urine sediment further confirms infection, where organisms may appear as cocci or bacilli alongside WBCs, strengthening diagnostic confidence. In immunocompromised patients or those under prolonged antibiotic treatment, yeast cells such as *Candida albicans* may also be observed, appearing as budding oval-shaped structures and indicating fungal UTIs [20]. Recent advancements in Artificial Intelligence, particularly deep learning models like CNNs and YOLO-based systems, have significantly improved automated detection of these elements, achieving high accuracy levels between 88% and 95% [11], [12], [3]. However, most existing approaches analyze each component separately, highlighting the need for integrated diagnostic systems that combine image analysis with clinical patient data for more reliable UTI detection [15], [19].

1.1.2 Kidney Stone Risk Prediction using Urinary Crystals

Urinary crystals play a crucial role in the early identification and prediction of kidney stone formation. These microscopic structures form due to the supersaturation of salts and minerals in urine and are strongly associated with metabolic imbalances, dehydration, dietary habits, and urinary tract conditions. The most clinically significant crystals include calcium oxalate, uric acid, struvite, and cystine crystals, each reflecting different underlying pathological mechanisms [21], [24]. Calcium oxalate crystals are the most commonly observed and are often linked to low fluid intake and high oxalate consumption, while uric acid crystals are associated with acidic urine and disorders such as gout. Struvite crystals typically indicate infection-related stones caused by urease-producing bacteria, and cystine crystals are rare but strongly indicative of genetic disorders such as cystinuria [10], [24].

From a diagnostic perspective, urinary crystals exhibit distinct morphological characteristics under microscopic examination, making them suitable for automated recognition using deep learning techniques. For example, calcium oxalate crystals often appear as envelope or dumbbell-shaped structures, uric acid crystals present as rhomboid or rosette forms, struvite crystals resemble “coffin-lid” shapes, and cystine crystals appear as hexagonal plates. These well-defined visual patterns enable machine learning and deep learning models, such as CNNs and object detection frameworks, to accurately classify crystal types from urine sediment images [3], [6]. By integrating crystal detection with clinical interpretation, AI-based systems can predict kidney stone risk at an early stage, improve diagnostic efficiency, and support clinicians in providing timely preventive treatment strategies [14], [15].

1.1.3 Hematuria Analysis using RBC Morphology

The presence of red blood cells (RBCs) in urine, known as hematuria, is an important clinical indicator of underlying urinary tract or renal disorders. Hematuria can result from a wide range of conditions, including urinary tract infections, kidney stones, trauma, and more serious diseases such as glomerulonephritis or malignancies [20], [22]. A key diagnostic factor in determining the origin of bleeding is the morphological analysis of RBCs. Isomorphic RBCs, which maintain a uniform shape, typically indicate non-glomerular bleeding sources such as the bladder or urethra. In contrast, dysmorphic RBCs, which exhibit irregular shapes and membrane distortion, are strongly associated with glomerular damage, particularly in conditions like glomerulonephritis [22], [23]. Therefore, RBC morphology provides essential diagnostic insight for differentiating between upper and lower urinary tract pathologies.

Recent advances in artificial intelligence have enabled the use of deep learning models, particularly convolutional neural networks (CNNs), to automate RBC classification in urine microscopy. These models are capable of learning complex morphological features and distinguishing between isomorphic and dysmorphic RBCs with improved consistency compared to manual analysis [6], [7]. However, accurate classification remains challenging due to variations in cell shape, overlapping cells, staining differences, and inconsistent imaging conditions. Despite these limitations, AI-based RBC analysis has the potential to significantly reduce diagnostic errors, enhance clinical efficiency, and support early detection of renal diseases by providing reliable and standardized hematuria assessment [11], [15].

1.1.4 Renal Disease Diagnosis using Urinary Casts

Urinary casts are cylindrical structures formed within the renal tubules and collecting ducts, primarily composed of Tamm-Horsfall protein, and sometimes containing cellular or granular elements. They serve as important diagnostic indicators of kidney health, as different cast types are closely associated with specific renal pathologies [2], [23]. RBC casts are strongly indicative of glomerulonephritis, reflecting glomerular bleeding and inflammation [22]. WBC casts are commonly linked to pyelonephritis and interstitial nephritis, suggesting active infection or inflammation within the kidney [2], [9]. Granular casts are typically associated with acute tubular necrosis (ATN), often resulting from ischemic or toxic injury, while waxy casts are seen in chronic kidney disease and indicate long-standing renal impairment [2], [23]. Therefore, the presence and type of urinary casts provide valuable insights into both the location and severity of kidney damage [4], [14].

Despite their diagnostic significance, accurate detection and classification of urinary casts remain challenging due to their diverse morphology, overlapping visual features, and variations in microscopic imaging quality [2], [4], [11]. Distinguishing between subtypes such as granular and waxy casts is particularly difficult, even for experienced observers [2], [23]. Recent advances in artificial intelligence, especially object detection and deep learning classification models, have shown promising improvements in automating cast identification [3], [6], [11], [12]. However, challenges such as class imbalance, limited annotated datasets, and subtle inter-class differences still affect model performance [5], [13], [18]. Integrating cast analysis with other urine sediment components and patient clinical data can enhance diagnostic accuracy, enabling a more comprehensive and reliable assessment of renal diseases [15], [19], [25].

1.2 Research Gap

1.2.1 Research Gap of UTI Detection

A major challenge in urinary tract infection (UTI) detection using urine microscopy is the fragmented nature of existing artificial intelligence approaches, where most models focus on detecting individual components such as white blood cells (WBCs), bacteria, or yeast in isolation rather than integrating them into a comprehensive diagnostic framework [3], [6], [11], [12]. Although these models report high accuracy in detection and classification tasks, they often fail to translate microscopic findings into clinically meaningful diagnostic outcomes, limiting their real-world applicability. In particular, current systems tend to operate either as particle-level detectors or as black-box prediction models based on symptoms, without incorporating clinically informed reasoning into the decision-making process.

Another significant limitation lies in bacterial analysis, where many deep learning approaches rely on culture-based datasets or classification techniques instead of direct detection from urine microscopy images [16], [17]. This creates a gap between research and clinical practice, as real-world urinalysis requires accurate localization and quantification of bacteria within microscopic samples. Furthermore, the lack of annotated datasets for bacteria detection restricts the development of robust detection and segmentation models, reducing the ability to assess infection severity and bacterial load.

In addition, existing methodologies do not effectively combine multiple microscopic indicators such as WBCs, bacteria, and yeast to differentiate between bacterial and fungal UTIs or to identify the severity and location of infection, including upper and lower UTIs [1], [9], [16], [17]. The absence of such integrated analysis limits the clinical usefulness of current systems, as accurate diagnosis depends on the interaction of multiple sediment components rather than isolated observations. Moreover, most models do not incorporate important clinical evidence such as WBC casts, which are essential for distinguishing conditions like pyelonephritis from lower urinary tract infections [2], [4], [23].

Another critical gap is the lack of integration between microscopy findings and patient metadata, including symptoms, medical history, and demographic factors. While some machine learning models use clinical data for UTI prediction, they often lack microscopic interpretability, whereas image-based models fail to incorporate contextual clinical information [16], [19]. This separation reduces diagnostic accuracy and limits the ability to provide personalized and clinically relevant predictions.

Furthermore, current systems face challenges in generalization due to limited and non-diverse datasets, as well as inconsistencies in image acquisition, annotation standards, and magnification levels across studies [6], [12], [18]. These issues hinder the development of robust models capable of performing reliably across different clinical environments. The computational complexity of deep learning models also limits their deployment in real-time and resource-constrained settings, reducing their practical applicability in routine laboratory workflows [25].

More importantly, existing approaches do not provide an end-to-end diagnostic framework that integrates detection, analysis, and clinical interpretation. There is no system that simultaneously analyzes WBCs, bacteria, and yeast and applies medically informed decision logic to classify different types of UTIs in a transparent and explainable manner. As a result, current models fail to deliver holistic disease prediction and clinical decision support.

To address these limitations, this research proposes an integrated AI-based framework that combines multi-component urine microscopy analysis with clinical metadata and rule-based reasoning to enable accurate, interpretable, and real-time UTI diagnosis. By shifting from isolated detection tasks to a unified diagnostic approach, the proposed system aims to improve clinical reliability, enhance interpretability, and support effective decision-making in real-world healthcare environments.

Table 1.1: Comparison of Existing Systems and Proposed System (UTI Detection)

Research	Bacterial Morphology Classification	WBC Cast Integration	Multimodal Integration (Image + Metadata)	Real-Time Clinical Use
Research A [4]	No	No	No	No
Research B [7]	No	Yes	No	No
Research C [12]	No	No	Yes	No
Research D [8]	No	No	No	Yes
Proposed System	Yes	Yes	Yes	Yes

1.2.2 Research Gap of Kidney Stone Risk Prediction

A major challenge in kidney stone risk prediction using urine microscopy is that existing artificial intelligence approaches primarily focus on the detection and classification of urinary crystals without effectively linking these findings to clinically meaningful risk assessment [5], [21], [24]. While deep learning models such as CNN-based classifiers and object detection frameworks have demonstrated high accuracy in identifying crystal types, including calcium oxalate, uric acid, and struvite, these systems largely operate as isolated classification tools rather than integrated diagnostic solutions. As a result, they fail to translate microscopic crystal characteristics into personalized predictions of kidney stone formation, limiting their clinical applicability.

Another significant limitation is the lack of quantitative and contextual analysis of crystals. Most current systems focus only on crystal presence or type, without considering important factors such as crystal count, size, aggregation patterns, and distribution within the urine sample. These factors are clinically important, as crystal density and clustering are strong indicators of stone formation risk. The absence of such quantitative analysis reduces the ability of existing models to provide accurate and reliable risk predictions.

Furthermore, many models struggle with distinguishing between visually similar crystal types due to overlapping morphological features, which leads to misclassification and reduced diagnostic accuracy [21], [24]. This issue is further compounded by dataset limitations, as most available datasets are relatively small and lack diversity in terms of crystal types, imaging conditions, and patient populations

[18]. Consequently, models trained on such datasets often fail to generalize effectively across different clinical environments.

Another critical gap is the lack of integration between microscopic findings and patient-specific clinical data. Factors such as hydration level, dietary habits, metabolic conditions, and medical history play a crucial role in kidney stone formation, yet most existing systems do not incorporate this information into their prediction models. While some studies have explored machine learning approaches for disease prediction, they often rely solely on clinical data without integrating image-based evidence, resulting in limited interpretability and reduced diagnostic accuracy [19].

Additionally, current approaches lack explainable AI mechanisms that can provide insights into how crystal features contribute to kidney stone risk. Most systems function as black boxes, offering predictions without clear reasoning, which reduces clinician trust and limits adoption in real-world healthcare settings. The computational complexity of deep learning models also poses challenges for real-time deployment, particularly in resource-constrained environments where efficient and lightweight solutions are required [25].

More importantly, there is no comprehensive system that integrates crystal detection, classification, quantitative analysis, and clinical interpretation into a unified framework for personalized kidney stone risk prediction. Existing models do not provide end-to-end decision support that combines microscopic findings with patient metadata to generate clinically actionable insights.

To address these limitations, this research proposes a multimodal AI-based framework that integrates urinary crystal detection with quantitative analysis and patient metadata to enable accurate, interpretable, and personalized kidney stone risk prediction. By shifting from isolated classification tasks to a clinically driven diagnostic approach, the proposed system aims to improve prediction accuracy, enhance interpretability, and support effective clinical decision-making in real-world settings.

Table 1.2 summarizes key research gaps from existing studies and highlights how this study addresses them through novel techniques and practical system design.

Table 1.2: Comparison of previous research of AMD

Research	**Early Detection**	**Stage Identification**	**Personalized medical records**
Research A [9]	✓	✗	✗
Research B [10]	✓	✓	✗
Research C [11]	✓	✓	✗
Proposed System	✓	✓	✓

1.2.3 Research Gap of Hematuria Analysis (RBC Morphology)

A major challenge in hematuria analysis using urine microscopy is the limited capability of existing artificial intelligence models to translate red blood cell (RBC) morphological findings into clinically meaningful diagnostic outcomes. Although several deep learning approaches have been developed to detect and classify RBCs in urine microscopy images, most studies primarily focus on improving classification accuracy rather than supporting clinical decision-making processes [10], [22]. These models typically differentiate between isomorphic and dysmorphic RBCs, but they often fail to utilize this information effectively to identify the underlying cause of hematuria, such as distinguishing between glomerular and non-glomerular bleeding.

Another significant limitation is the variability in RBC morphology, which makes accurate classification challenging. Dysmorphic RBCs, which are critical indicators of glomerular diseases, exhibit irregular shapes and overlapping structures, leading to reduced detection accuracy and recall in many models. This issue is further compounded by limited and non-diverse datasets, which restrict the ability of models to generalize across different imaging conditions and patient populations [18]. As a result, existing systems often struggle to maintain consistent performance in real-world clinical environments.

Furthermore, most current approaches analyze RBCs in isolation without integrating other relevant urine sediment components such as WBCs, casts, or crystals. In clinical practice, hematuria diagnosis often depends on the combination of multiple indicators.

For example, the presence of dysmorphic RBCs along with RBC casts strongly suggests glomerulonephritis, yet existing systems do not incorporate such relationships into their diagnostic frameworks [2], [23]. This lack of integration limits the ability of current models to provide comprehensive and accurate diagnostic insights.

Another critical gap is the absence of multimodal integration with patient metadata. Factors such as patient history, trauma, infections, and lifestyle play an essential role in determining the cause of hematuria. However, most image-based models do not incorporate this contextual information, while clinical prediction models lack microscopic interpretability. This separation reduces diagnostic accuracy and limits the clinical usefulness of existing approaches [19].

Additionally, explainability remains a major concern in current hematuria detection systems. Most deep learning models function as black boxes, providing predictions without clear reasoning or visual justification. This lack of transparency reduces clinician trust and hinders the adoption of AI-based systems in healthcare settings. Moreover, computational complexity and the need for high-quality annotated datasets further limit the deployment of these models in real-time clinical environments [25].

More importantly, there is no comprehensive system that integrates RBC detection, morphological classification, and clinical reasoning into an end-to-end framework for hematuria cause prediction. Existing systems fail to combine microscopic evidence with clinical context to generate interpretable and actionable diagnostic outcomes.

To address these limitations, this research proposes a multimodal AI-based framework that integrates RBC morphology analysis with other urine sediment components and patient metadata to enable accurate, interpretable, and clinically relevant hematuria diagnosis. By moving beyond isolated classification tasks toward a unified diagnostic approach, the proposed system aims to improve diagnostic accuracy, enhance interpretability, and support effective clinical decision-making in real-world healthcare environments.

Table 1.3: Comparison of previous research of Glaucoma

Research	**Early Detection**	**Stage Identification**	**Explainable AI**	**Hybrid Approach**

Research A [16]	✓	✗	✗	✗
Research B [18]	✓	✗	✗	✓
Research C [55]	✓	✓	✗	✗
Research D [56]	✓	✗	✓	✗
Proposed System	✓	✓	✓	✓

1.2.4 Research Gap of Renal Disease Diagnosis (Urinary Casts)

A major challenge in urinary cast analysis for renal disease diagnosis is the limited focus of existing artificial intelligence approaches on basic detection rather than comprehensive subtype classification and clinical interpretation. While several deep learning models have been developed to detect urinary casts from microscopy images, these approaches often treat cast detection as an isolated task and do not extend their analysis to clinically relevant subtype identification, such as hyaline, granular, red blood cell (RBC), white blood cell (WBC), and waxy casts [2], [4]. This limitation reduces their usefulness in real-world diagnostic settings, where different cast types are strongly associated with specific renal conditions.

Another significant gap is the difficulty in accurately classifying rare and visually similar cast subtypes. Many datasets used in existing studies suffer from class imbalance, with certain cast types, such as RBC and waxy casts, being underrepresented. This leads to poor model performance in detecting clinically critical but less frequent cast categories, ultimately affecting diagnostic reliability [23]. In addition, the complex morphology and overlapping structures of casts further complicate segmentation and classification tasks, resulting in reduced accuracy under real-world imaging conditions.

Furthermore, most current systems do not integrate cast analysis with other urine sediment components such as RBCs, WBCs, or crystals. In clinical practice, renal disease diagnosis relies on the combination of multiple findings. For example, the presence of RBC casts along with dysmorphic RBCs is a strong indicator of glomerulonephritis, while WBC casts are associated with pyelonephritis [2], [23]. However, existing models fail to incorporate such relationships into a unified diagnostic framework, limiting their ability to provide comprehensive clinical insights.

Another critical limitation is the lack of integration with patient metadata. Important clinical factors such as medical history, symptoms, and laboratory findings are rarely combined with microscopic analysis in current systems. While some studies explore either image-based detection or clinical prediction models independently, the absence of multimodal integration reduces the overall diagnostic accuracy and personalization of predictions [19].

Additionally, explainability remains a key concern in cast detection systems. Most deep learning models operate as black boxes, offering predictions without clear reasoning or visualization of features contributing to the diagnosis. This lack of transparency reduces clinician trust and limits the adoption of such systems in healthcare environments. Moreover, computational complexity and the requirement for high-quality annotated datasets further restrict the deployment of these models in real-time and resource-constrained clinical settings [25].

More importantly, there is no comprehensive system that integrates cast detection, subtype classification, and clinical reasoning into an end-to-end framework for renal disease diagnosis. Existing approaches fail to combine microscopic evidence with contextual clinical data to generate interpretable and actionable diagnostic outcomes.

To address these limitations, this research proposes a multimodal AI-based framework that integrates urinary cast detection and classification with other urine sediment components and patient metadata to enable accurate, interpretable, and clinically relevant renal disease diagnosis. By transitioning from isolated detection tasks to a unified diagnostic approach, the proposed system aims to improve accuracy, enhance interpretability, and support effective clinical decision-making in real-world healthcare environments.

The below Table 1.4 shows a tabularized format of the above explanation with regard to the diagnosis of Retinal Vein Occlusion (RVO).

Table 1.4: Comparison of former researches

Application reference	**Early Detection**	**Progression Monitoring**	**All-Type Identification**
Research A	✓	X	X
Research B	✓	✓	X
Research C	✓	X	X
Proposed System	✓	✓	✓

1.3 Research Problem

Visual impairments caused by retinal diseases pose a major public health burden worldwide, with conditions such as glaucoma, diabetic retinopathy (DR), age-related macular degeneration (AMD), and retinal vein occlusion (RVO) ranking among the leading causes of preventable blindness. Early detection of these conditions is critical for initiating treatment during the stages when intervention is most effective. However, in many regions, particularly in low- and middle-income countries, there is a shortage of trained ophthalmologists and access to regular eye screenings, resulting in delayed diagnoses and severe vision loss. Even in well-equipped clinical settings, the manual assessment of retinal images can be labor-intensive, prone to inter-observer variability, and inefficient for large-scale screening.

The emergence of deep learning in the medical imaging field has opened up new possibilities for automated disease detection. Convolutional Neural Networks (CNNs), Vision Transformers, and hybrid models have demonstrated impressive performance in analyzing complex visual patterns in fundus images. Despite these advancements, there remains a critical knowledge gap in identifying which architectures are most effective for detecting specific retinal diseases. For instance, the morphological features indicative of glaucoma differ significantly from those of DR or AMD, necessitating tailored model strategies or highly generalizable frameworks capable of multi-class classification with clinical-grade accuracy.

Another challenge in this domain is the availability and quality of annotated datasets. Retinal disease datasets often exhibit class imbalance, varying image resolutions, and inconsistent labeling, all of which hinder the performance of deep learning models. Moreover, the overlapping symptoms across diseases further complicate the classification task, increasing the likelihood of false positives or missed diagnoses. Consequently, there is an urgent need to explore which architectural choices such as depth, feature extraction mechanisms, attention layers, or ensemble techniques can best handle these complexities while maintaining robustness and interpretability.

Furthermore, the deployment of deep learning-based systems in real-world settings requires not only accuracy but also efficiency and generalizability across diverse populations. A model trained on a specific dataset may not generalize well to images captured under different conditions or from different demographic groups. Therefore, in addition to evaluating detection accuracy, it is essential to consider how adaptable and scalable these models are when integrated into clinical workflows. Investigating this can provide insights into designing practical and ethical AI-powered tools for ophthalmic diagnostics.

1.3.1 Research Problem Statement

What are the most effective deep learning architectures for detecting specific retinal diseases such as glaucoma, diabetic retinopathy (DR), age-related macular degeneration (AMD), and retinal vein occlusion (RVO) from fundus images?

1.4 Research Objectives

The objective of this research is to develop an integrated, AI-powered diagnostic system for urine microscopy analysis that improves the accuracy, efficiency, and clinical relevance of diagnosing urinary tract and renal diseases. The study focuses on addressing the limitations of existing fragmented systems by combining multiple urine sediment components with clinical data into a unified framework.

The specific objectives of this research include:

- To design and implement deep learning models to detect and classify key urine sediment components such as red blood cells (RBCs), white blood cells (WBCs), bacteria, yeast, urinary crystals, and casts from microscopy images.

- To analyze RBC morphology and differentiate between isomorphic and dysmorphic RBCs in order to identify glomerular and non-glomerular causes of hematuria.

- To develop a UTI prediction mechanism by integrating WBC, bacteria, and yeast detection with clinically informed decision rules to classify different types of infections.

- To detect and classify urinary crystals and utilize their characteristics to predict the risk of kidney stone formation.

- To identify and classify urinary casts and use them for diagnosing renal diseases such as glomerulonephritis, pyelonephritis, and chronic kidney disease.

- To integrate patient metadata, including age, symptoms, medical history, and lifestyle factors, with image-based findings using a multimodal approach to improve diagnostic accuracy and personalization.

- To incorporate explainable AI techniques that provide transparent and interpretable diagnostic outputs to enhance clinician trust and usability.

➢ To develop a real-time, web-based system that enables users to upload microscopy images and receive automated diagnostic predictions for practical clinical use.

2. METHODOLOGY

2.1 UroAI System Architecture Overview

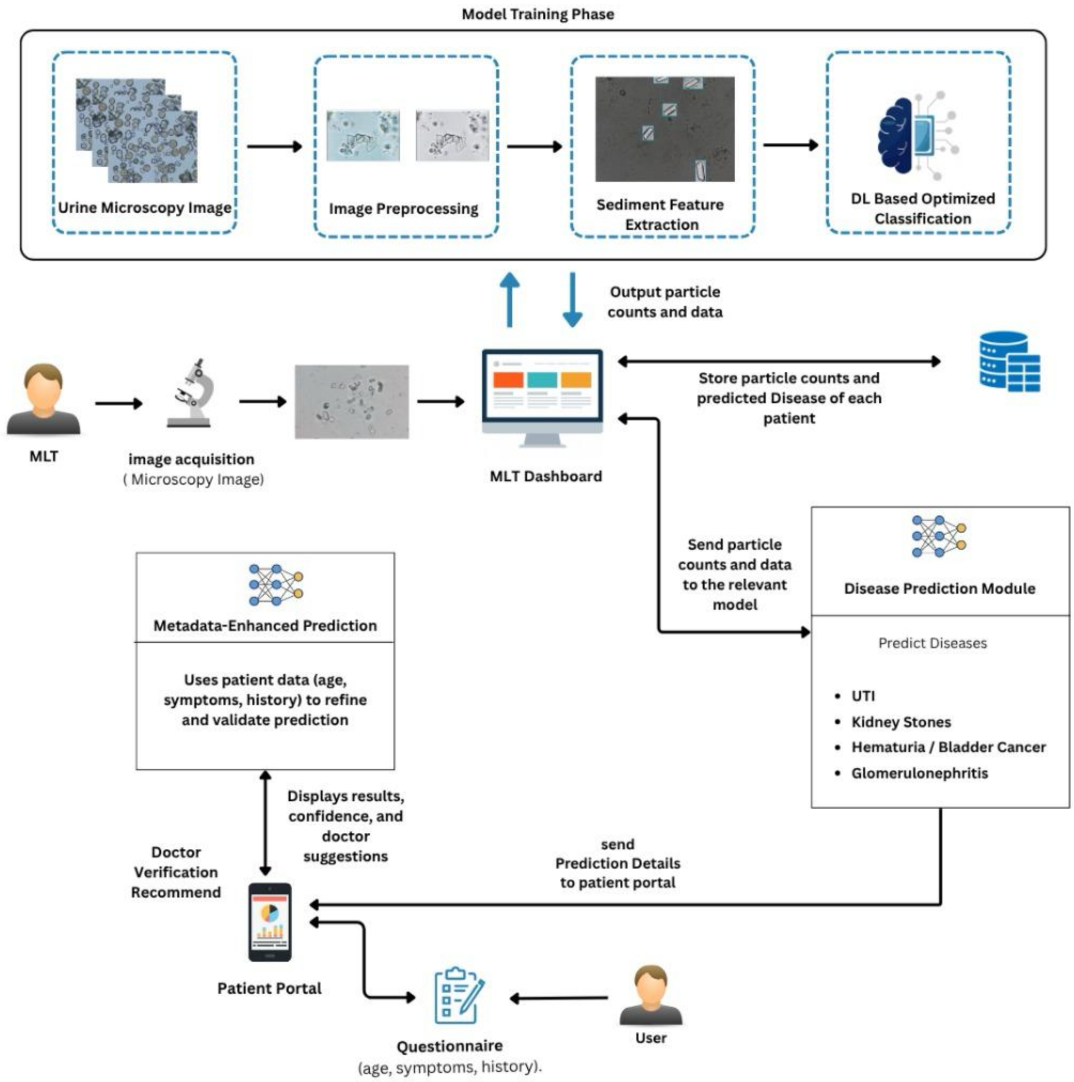

Figure 2.1: Overall architectural diagram

To effectively implement the proposed AI-powered diagnostic system for urinary sediment analysis and disease prediction, a robust and modular methodology has been adopted, covering the development, training, and deployment of deep learning models capable of detecting and classifying key urinary particles such as RBCs, WBCs, casts, crystals, and bacteria from highresolution microscopy images. The system also integrates structured patient metadata such as age, symptoms, lifestyle habits, and medical history through multimodal learning to enhance diagnostic accuracy and personalize disease predictions. The methodology follows a structured pipeline comprising data acquisition, image preprocessing, model development, metadata fusion, validation, and deployment via a real-time web interface. This comprehensive approach ensures clinical reliability, scalability, and usability in real-world laboratory environments while supporting the early detection of conditions such as UTIs, kidney stones, glomerulonephritis, and bladder cancer through explainable AI-driven insights.

The architecture of the proposed system is illustrated in **Figure 2.1** and comprises a modular and scalable pipeline that automates the process of urinary sediment analysis, metadata integration, and disease prediction. It begins with the acquisition of high-resolution urine microscopy images using laboratory-grade microscopes equipped with digital imaging sensors. Medical Laboratory Technicians (MLTs) capture images representing key urinary sediment particles such as red blood cells (RBCs), white blood cells (WBCs), urinary casts, crystals and bacteria. These images are then subjected to a preprocessing stage where techniques such as contrast enhancement, histogram equalization, denoising, and resizing are applied to normalize image quality across samples, minimize device-induced inconsistencies, and prepare the data for model input.

Once preprocessing is complete, sediment feature extraction is carried out using computer vision and deep learning techniques to isolate morphological attributes such as particle shape, size, texture, boundary definition. The extracted features are fed into a deep learning-based classification module composed of YOLOv5 and hybrid CNN architectures, which are trained to accurately detect and categorize particles into clinically meaningful groups. These include isomorphic and dysmorphic RBCs for hematuria analysis, four cast types (RBC, WBC, granular, and waxy) for kidney condition assessment, crystal subtypes (e.g., calcium oxalate, uric acid, struvite, cystine) for kidney stone risk estimation, and WBCs or bacterial forms (e.g., cocci, bacilli) for UTI prediction. Each particle type is identified and counted per high-power field (HPF) and stored for diagnostic inference.

Following classification, an initial diagnostic report is automatically generated and displayed on the MLT interface. This report includes preliminary predictions for conditions such as urinary tract infections, kidney stones, glomerulonephritis, and hematuria-related cancers, including the prediction of hematuria causes by differentiating between glomerular and nonglomerular bleeding based on RBC morphology. Importantly, no patient metadata is required at this stage. Instead, the system prompts the patient to access their report through a secure web portal and encourages them to submit personal health information such as age, gender, urinary symptoms, hydration status, and relevant medical history via a structured form. This patientdriven metadata submission model reduces the burden on MLTs while improving prediction specificity.

Once the metadata is received, it is combined with the image-based features using a multimodal fusion model. This model employs a late fusion strategy enhanced with attention mechanisms to generate more personalized and accurate predictions. For example, it enables the system to distinguish glomerular versus non-glomerular hematuria based not only on RBC morphology but also on risk factors such as trauma or smoking history. The refined report, including updated disease probabilities, confidence scores, and personalized suggestions, is then made available to the patient through their dashboard. If any critical indicators are detected, the system can recommend further doctor review or medical follow-up.

2.2 Detailed Infrastructure and Service Flow of UroAI

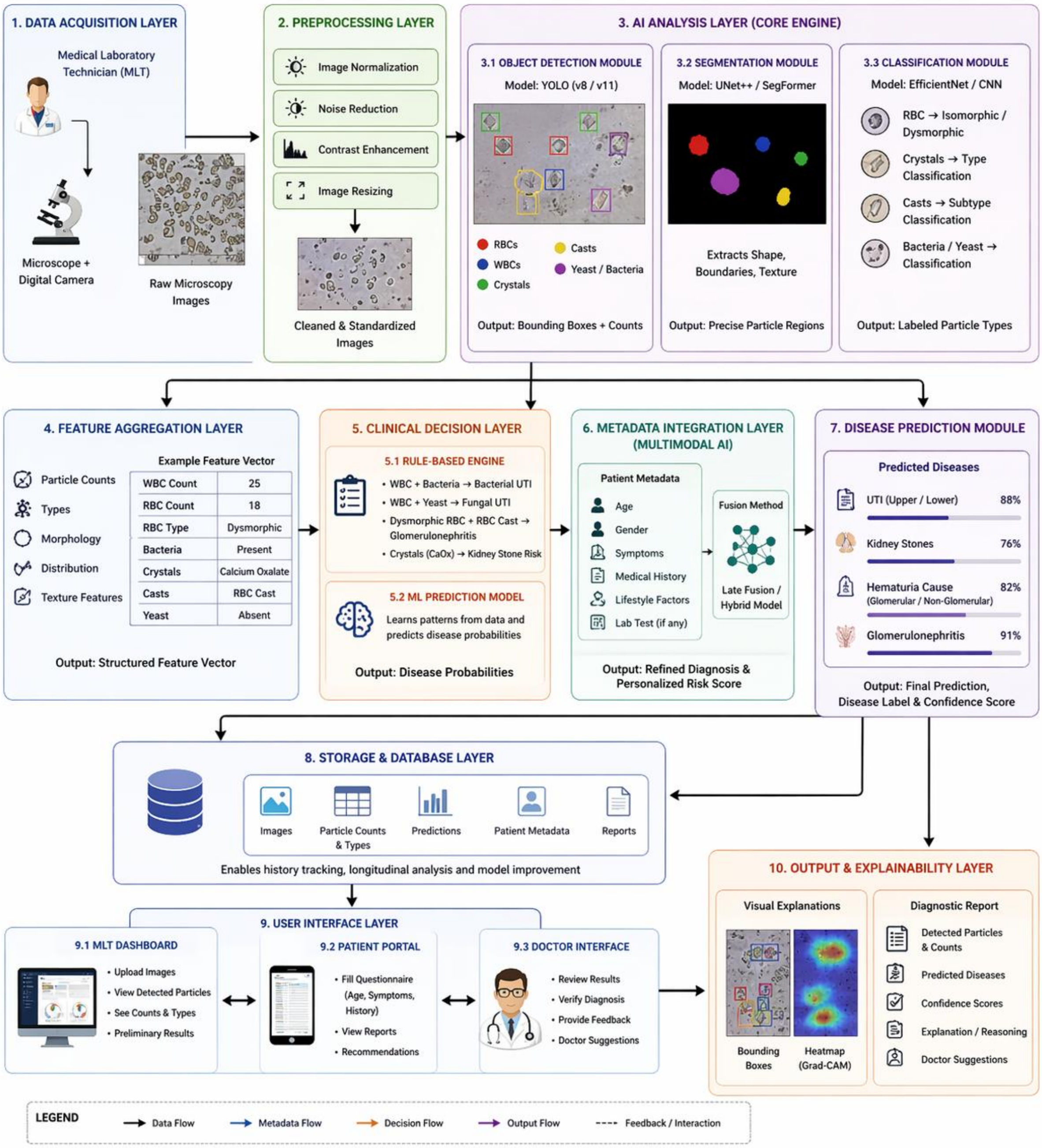

Figure 2.2: Detailed System Diagram

2.2.1 User Interface and Access Control

The proposed system implements a web-based user interface designed using modern frontend technologies to ensure responsiveness, usability, and seamless interaction across different user roles. The frontend of the system is developed using **React.js** with **Tailwind CSS**, enabling a clean, responsive, and component-based UI design. This allows efficient rendering of dynamic data such as detected particles, diagnostic results, and visual outputs including bounding boxes and segmentation overlays.

The system supports three primary user roles: **Medical Laboratory Technician (MLT), Doctor, and Patient**, each interacting with the platform through dedicated interfaces. The MLT dashboard enables image upload, visualization of detected urine sediment components (RBCs, WBCs, bacteria, yeast, crystals, and casts), and access to preliminary analysis results. The Doctor interface provides advanced diagnostic views, including disease predictions, confidence scores, and explainable AI outputs such as heatmaps, allowing clinicians to validate and interpret system predictions. The Patient portal allows users to input clinical information through a structured questionnaire and view their diagnostic reports and recommendations in an understandable format.

For backend communication, the frontend interacts with a RESTful API built using **FastAPI (Python)**, which handles image processing requests, model inference, and data retrieval. The system ensures smooth data flow between the frontend and AI modules, enabling real-time response and efficient processing.

Access control is implemented using a **role-based access control (RBAC)** mechanism to ensure secure and authorized usage of the system. User authentication is managed using Firebase Authentication, which supports secure login, session management, and identity verification. Each user role is assigned specific permissions, ensuring that users can only access data relevant to their role. For example, patients can only view their own reports, while doctors can access patient diagnostic results for clinical validation, and MLTs are restricted to data processing and image analysis functionalities.

Data security is further enhanced through **token-based authentication (JWT)** and encrypted communication between the frontend and backend services. The system also enforces validation and access restrictions at the API level to prevent unauthorized data access.

Overall, the combination of React-based UI, FastAPI backend services, and Firebase authentication provides a secure, scalable, and user-friendly interface that supports efficient interaction and maintains the confidentiality and integrity of medical data.

2.2.2 Backend Processing and Image Management

The backend of the system is implemented using **FastAPI (Python)** to handle image processing, model inference, and communication between system components. When a urine microscopy image is uploaded, the backend performs preprocessing steps such as resizing, normalization, and format conversion before passing the image to the AI models. The backend manages a modular processing pipeline where detection (YOLO), segmentation, and classification models operate sequentially to identify and analyze urine sediment components. The outputs from these models are combined to generate structured results, including particle types, counts, and features, which are then used for disease prediction.

For image management, the system stores both original and processed images, including annotated outputs with detection results. Each image is linked with its corresponding analysis and metadata, enabling efficient retrieval, visualization, and tracking. The backend ensures efficient handling of image data and supports real-time processing for practical clinical use.

2.2.3 AI Integration through API Gateway

The AI components are integrated through an API gateway implemented within the backend using **FastAPI**, which acts as a central entry point for handling all processing requests. The API gateway manages incoming image and metadata requests, routes them to the appropriate AI models, and coordinates the overall processing pipeline.

Once an image is received, the API gateway directs it to detection, segmentation, and classification modules, and collects their outputs. These results are then passed to the decision-making layer to generate final predictions. This centralized approach ensures efficient communication between system components, simplifies model integration, and allows easy updates or scaling of individual AI modules. Additionally, the API gateway improves system performance and security by controlling request flow, validating inputs, and ensuring that all processing is handled consistently within the backend environment.

2.2.4 Prediction Response and Visualization

The system generates prediction responses based on the combined outputs of detection, classification, and decision-making modules. After processing the microscopy image and integrating relevant features, the backend produces diagnostic results, including identified particle types, counts, and predicted diseases such as UTI, kidney stone risk, hematuria cause, or renal conditions. Each prediction is accompanied by a confidence score to indicate the reliability of the result.

For visualization, the system provides annotated images with bounding boxes and segmentation overlays to highlight detected particles such as RBCs, WBCs, bacteria, yeast, crystals, and casts. In addition, explainable AI techniques such as heatmaps or feature highlighting are used to show the regions that contributed to the prediction. The final output is presented through the user interface as a structured report, allowing users to clearly understand both the results and the reasoning behind them.

2.2.5 Security, Modularity, and Cloud Optimization

The system ensures security through role-based access control (RBAC) and token-based authentication, allowing only authorized users to access specific functionalities and data. Secure communication protocols are used to protect data during transmission, and validation mechanisms are applied at the backend to prevent unauthorized access and data misuse. The architecture is designed to be modular, where different components such as detection, classification, and prediction modules operate independently. This modularity allows easy updates, maintenance, and scalability, enabling new models or features to be integrated without affecting the overall system.

For cloud optimization, the system is designed to support deployment on cloud platforms, enabling scalable storage and processing of microscopy images. Efficient data handling, lightweight models, and optimized backend processing ensure faster response times and reduced computational overhead, making the system suitable for both high-resource and resource-constrained environments.

2.3 Main Diagnosis Components of UroAI

2.3.1 UTI Component

2.3.1.1 Data Collection

The data collection process for the UTI component in the proposed **UroAI system** focuses on acquiring high-quality urine microscopy images along with relevant clinical metadata to enable accurate infection detection and classification. Urine samples are obtained from laboratory settings and captured using **digital microscopy systems**, where Medical Laboratory Technicians (MLTs) acquire high-resolution images representing key urinary sediment particles such as **white blood cells (WBCs), bacteria, and yeast cells**. These particles are essential biomarkers used in identifying urinary tract infections. The collected dataset is organized into distinct categories based on particle types. Separate annotated datasets are prepared for WBCs, yeast cells, and bacterial presence to support specialized model training. Annotation includes bounding boxes for detection tasks and pixel-level masks for segmentation tasks, particularly for WBCs and yeast cells.

Due to the limited availability of precise annotations for bacteria in microscopy images, bacterial data is prepared in the form of **cropped image samples** and labeled for classification tasks. These samples primarily focus on identifying common UTI-causing organisms such as *Escherichia coli*. In addition to image data, the system incorporates **structured patient metadata**, including age, symptoms (e.g., dysuria, urgency, fever), medical history, and lifestyle factors. This metadata is collected through a structured interface and used to enhance the diagnostic capability of the system by providing clinical context. If access to real clinical datasets is limited, **synthetic metadata generation techniques** are applied based on domain knowledge to simulate realistic patient profiles for training purposes. Overall, the data collection strategy ensures a **multimodal dataset** combining both image-based and clinical information, enabling the UroAI system to perform accurate, scalable, and clinically meaningful UTI diagnosis.

2.3.1.2. Selection of Best Model Architecture for UTI

The selection of the most suitable model architecture for the UTI component in the proposed **UroAI system** is based on the need to accurately detect, segment, and classify multiple urine sediment particles such as white blood cells (WBCs), bacteria, and yeast cells under varying microscopic conditions. For WBC and yeast detection, **YOLOv11-n** is selected as the primary object detection model due to its strong performance in **small object detection and real-time processing**. Urine microscopy images often contain dense and overlapping particles, making YOLOv11-n suitable for identifying multiple instances efficiently with high precision. To enhance morphological analysis, segmentation models are incorporated. A **UNet++ architecture with an EfficientNet-B0 encoder** is used for WBC segmentation, enabling accurate boundary detection and feature extraction at the pixel level. Similarly, yeast segmentation is performed using **SegFormer-B2**, which improves robustness in complex and dense microscopic regions.

Due to the lack of reliable bounding box annotations for bacteria, a detection-based approach is not feasible. Therefore, an **EfficientNet-B0 classification model** is selected for bacterial identification using cropped image samples. This approach allows effective classification of bacterial presence, particularly for common pathogens such as *Escherichia coli*. The combination of detection, segmentation, and classification models forms a **hybrid modular architecture**, where each model is optimized for a specific task. This modular design improves overall system performance and allows independent updates or enhancements to each component. Finally, a **rule-based decision layer** is integrated to combine outputs from all models and generate clinically meaningful UTI classifications. This ensures that model predictions are not only accurate but also interpretable and aligned with medical diagnostic logic. Overall, the selected architecture balances **accuracy, efficiency, and scalability**, making it suitable for real-time clinical deployment and robust UTI diagnosis.

2.3.1.3. Final Model Architecture Overview for UTI Detection

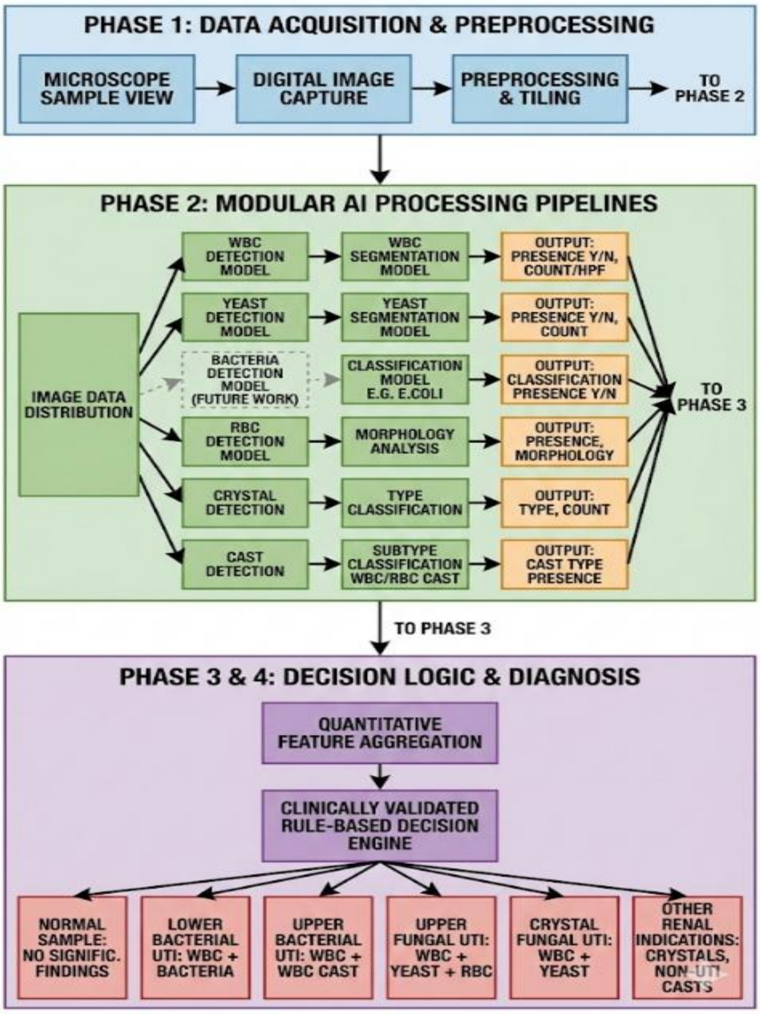

Figure 2.3: Final Model Architecture for UTI

As shown in **Figure 2.3**, the finalized model architecture for the **UTI component of the UroAI system** is designed as a modular and clinically driven deep learning pipeline that integrates multiple specialized models to analyze urine microscopy images. The process begins with data acquisition, where urine samples are captured using digital microscopy. These images then undergo preprocessing steps such as normalization, resizing, and tiling to ensure consistency and improve model performance. This preprocessing stage helps enhance image quality and prepares the data for accurate detection and analysis in the subsequent stages.

The core of the system is the **modular AI processing pipeline**, where different deep learning models are applied to detect and analyze specific urine sediment components. WBC and yeast detection are performed using YOLO-based models, followed by segmentation using architectures such as UNet++ and SegFormer to extract detailed morphological features. Due to annotation limitations, bacterial analysis is handled using a classification model (EfficientNet-B0). Each module produces structured outputs such as presence, count, and morphological features of particles. This modular design allows each component to be optimized independently while ensuring accurate and efficient analysis of complex microscopic patterns.

In the final stage, all extracted features are aggregated and passed into a **clinically validated rule-based decision engine**, which combines particle-level evidence to generate meaningful diagnostic outcomes. Based on the presence and combination of WBCs, bacteria, yeast, and other supporting features, the system classifies different UTI conditions such as lower bacterial UTI, upper bacterial UTI, fungal infections, or normal cases. This integration of deep learning outputs with clinical reasoning enables the system to move beyond simple detection and provide interpretable, real-world diagnostic predictions, making it a robust and scalable solution for automated urinalysis.

2.3.2 Kidney Stone Risk Prediction Component

2.3.2.1 Data Collection

The data collection process for the kidney stone component in the proposed **UroAI system** focuses on acquiring high-quality urine microscopy images containing urinary crystals, which are key indicators for kidney stone formation. The dataset is constructed using publicly available microscopy image sources and curated datasets that include common crystal types such as **calcium oxalate monohydrate, calcium oxalate dihydrate, uric acid, and phosphate crystals**. These crystal types are clinically significant as they are directly associated with different types of kidney stones.

The collected images are divided into training, validation, and testing sets using an **80:10:10 ratio** to ensure proper model evaluation and generalization. Each image is annotated with bounding boxes to identify crystal locations for the detection phase, while cropped crystal regions are labeled according to their specific types for classification tasks. To improve robustness and reduce overfitting, data augmentation techniques such as rotation, flipping, and scaling are applied to increase dataset diversity.

In addition to image data, the system also considers the **frequency and distribution of detected crystals**, which are important factors in assessing kidney stone risk. By combining crystal-level image data with quantitative information such as counts and types, the dataset supports both detection and personalized risk prediction within the system.

2.3.2.2 **Selection of model architecture**

The kidney stone component follows a **two-stage deep learning architecture** designed to balance detection accuracy and computational efficiency. In the first stage, a **YOLOv11m object detection model** is selected for crystal localization due to its strong performance in detecting small objects within complex microscopy images. The model uses pretrained weights and is trained using optimized parameters such as the AdamW optimizer, cosine learning rate scheduling, and early stopping to ensure stable convergence and high accuracy.

In the second stage, the detected crystal regions are cropped and passed into a classification model. An **EfficientNet-B0 convolutional neural network** is selected for crystal classification due to its ability to achieve high accuracy with low computational cost. The model is initialized with ImageNet pretrained weights and enhanced with layers such as global average pooling, batch normalization, and dropout to improve generalization and prevent overfitting. A softmax classifier is used to categorize crystal types into clinically relevant classes.

The combination of YOLO-based detection and EfficientNet-based classification forms a **hybrid pipeline**, where detection ensures precise localization and classification provides detailed subtype identification. This architecture is optimized for real-time performance and scalability, making it suitable for clinical deployment.

2.3.2.3 Final Model Architecture Overview for Kidney Stone Detection

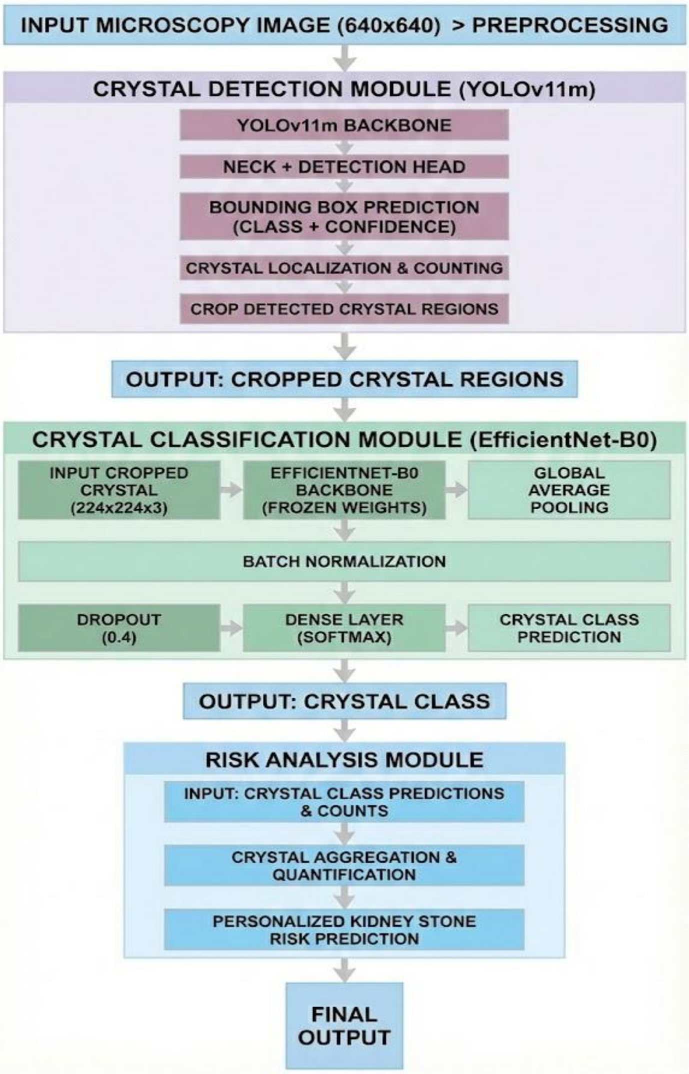

Figure 2.4: Final Model Architecture for Kidney Stone

As shown in **Figure 2.4**, the finalized model architecture for the **Kidney Stone Component of the UroAI system** is designed as a two-stage deep learning pipeline that integrates crystal detection, classification, and risk analysis to provide personalized kidney stone predictions. The process begins with input urine microscopy images, which are resized to a standardized resolution (640×640) and passed through a preprocessing stage. This stage ensures image consistency and improves the model's ability to detect small crystal structures under varying microscopic conditions.

The first stage of the pipeline focuses on **crystal detection using a YOLOv11m model**, which is selected for its strong performance in small object detection and real-time processing. The model utilizes a backbone and detection head to generate bounding box predictions along with class probabilities and confidence scores. Detected crystal regions are then localized and cropped to remove unnecessary background information. These cropped regions are passed to the second stage, where **EfficientNet-B0** is used for crystal classification. The classification model employs a pretrained backbone, followed by global average pooling, batch normalization, dropout, and a softmax layer to accurately classify crystal types such as calcium oxalate and uric acid.

In the final stage, the outputs from the classification model are aggregated and processed through a **risk analysis module**, which considers both the type and quantity of detected crystals. This module generates a personalized kidney stone risk prediction by analyzing the distribution and frequency of crystal types. The integration of detection, classification, and clinical interpretation enables the system to move beyond simple image analysis and provide meaningful diagnostic insights. This architecture ensures high accuracy, efficiency, and scalability, making it suitable for real-world clinical deployment.

2.3.3 Hematuria Component (RBC Morphology Prediction)

2.3.3.1 Data Collection

The data collection process for the hematuria component in the proposed **UroAI system** focuses on acquiring urine microscopy images containing red blood cells (RBCs), which are critical indicators for identifying the presence and cause of hematuria. The dataset consists of microscopic images collected from laboratory environments and publicly available sources, ensuring the inclusion of diverse RBC appearances under different imaging conditions. From these images, individual RBC instances are extracted using detection models, and approximately **2,000 single-RBC samples** are prepared for further analysis. These samples are manually verified to ensure quality by removing overlapping, damaged, or unclear cells. The dataset is then divided into training, validation, and testing sets to ensure proper evaluation.

Each RBC sample is labeled into two clinically significant categories: **isomorphic RBCs** (indicating non-glomerular causes) and **dysmorphic RBCs** (indicating glomerular causes). This classification is essential for identifying the underlying source of hematuria. Additionally, preprocessing techniques such as normalization and data augmentation (rotation and flipping) are applied to improve model robustness and generalization.

2.3.3.2 **Selection of Best Model Architecture**

The hematuria component adopts a **two-stage deep learning architecture** consisting of detection and classification phases. In the first stage, a **YOLOv8 object detection model** is used to accurately localize RBCs within microscopy images. YOLOv8 is selected due to its ability to efficiently detect small and densely distributed objects, making it suitable for RBC localization. Once RBCs are detected, individual cells are cropped and passed to the second stage, which focuses on morphological classification. An **EfficientNet-B3 convolutional neural network** is selected for this task due to its balance between performance and computational efficiency. The model is initialized with pretrained ImageNet weights and fine-tuned specifically for binary classification of RBC morphology.

The classification model analyzes structural features such as shape irregularities, membrane distortion, and size variations to distinguish between isomorphic and dysmorphic RBCs. This two-stage architecture ensures precise localization and accurate classification, enabling reliable clinical interpretation.

Table 2.3: Accuracy results of experimented models

Model	**Test Accuracy**	**Macro Precision**	**Macro Recall**
EfficientNet-B0	62.89%	0.65	0.63
EfficientNet-B2	73.20%	0.75	0.73
EfficientNet-B3	81.40%	0.82	0.81

The experimental results demonstrate that the selected architecture achieves strong performance in RBC morphology classification. The **EfficientNet-B3 model achieved a test accuracy of 81.40%**, with improved precision and recall compared to shallower models such as EfficientNet-B0 and B2. The use of YOLO-based detection significantly improved classification performance by isolating individual RBCs and reducing background noise. The model showed good sensitivity in detecting dysmorphic RBCs, which are clinically important for identifying glomerular diseases. However, some challenges remain due to visual similarities between RBC types and dataset limitations.

Overall, the hematuria component provides a reliable and clinically meaningful approach for distinguishing the source of blood in urine, supporting early diagnosis and improved decision-making in renal disease assessment.

2.3.3.3 **Final Model Architecture Overview for RBC Morphology Prediction**

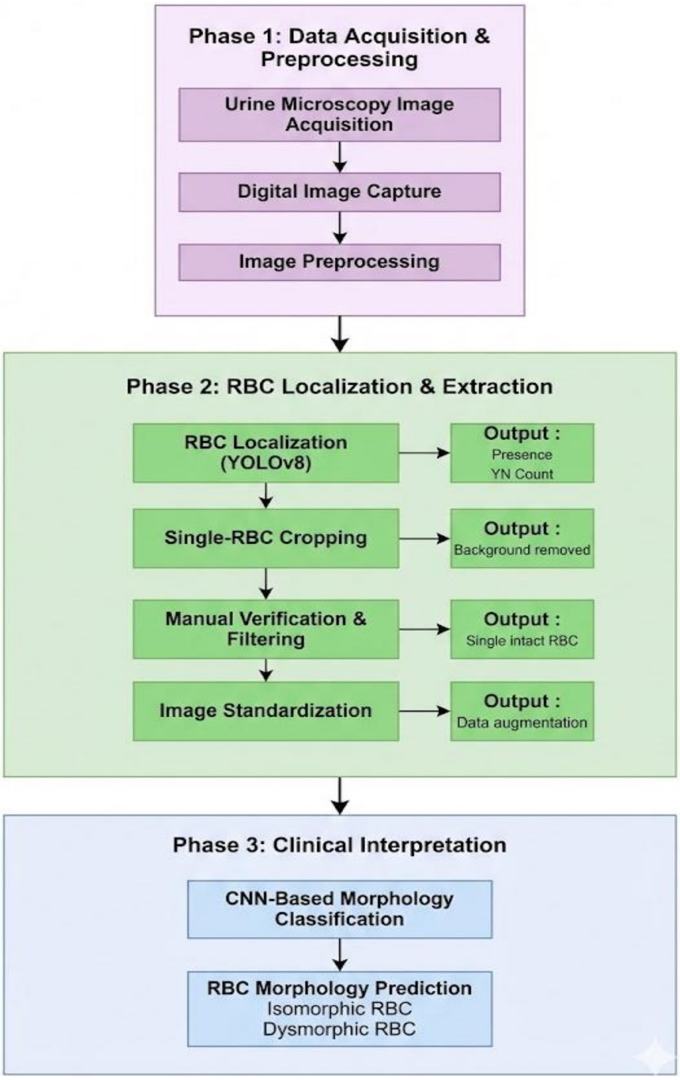

Figure 2.5: Model Architecture for Hematuria Component

As shown in **Figure 2.5**, the architecture of the hematuria component is designed as a structured multi-stage pipeline for RBC morphology analysis. The process begins with urine microscopy image acquisition and preprocessing, where images are standardized for consistent analysis. In the second phase, RBC localization is performed using the YOLOv8 model, followed by cropping of individual RBCs. A filtering process ensures that only high-quality, single RBC images are retained for further analysis. These images are then standardized and augmented to improve model performance.

In the final phase, the processed RBC images are passed into a CNN-based classification model, where morphological analysis is performed to distinguish between isomorphic and dysmorphic RBCs. The final output provides clinically relevant predictions that help differentiate between glomerular and non-glomerular causes of hematuria.

2.3.4 Cast Component (Renal Disease Diagnosis using Urinary Casts)

2.3.4.1 Data Collection and Preprocessing

The data collection process for the urinary cast component in the proposed **UroAI system** is based on a curated dataset developed using a structured pipeline through the Roboflow platform. The dataset consists of **5,391 urine microscopy images**, containing various sediment particles such as urinary casts, red blood cells, white blood cells, crystals, and other debris. This diversity ensures that the model learns to identify casts under realistic and complex microscopic conditions. The dataset is divided into training, validation, and testing sets using an **80:10:10 ratio** to ensure proper model generalization. Using the trained detection model, **1,436 images containing only cast regions** are extracted for further classification. These images are manually labeled into five clinically significant cast subtypes: **granular casts (338), hyaline casts (320), RBC casts (66), WBC casts (182), and waxy casts (22)**.

To address class imbalance, especially for rare subtypes such as waxy casts, targeted data augmentation techniques including rotation, brightness adjustment, flipping, and Gaussian noise injection are applied. These techniques improve dataset diversity and enhance model robustness. Additionally, future improvements such as GAN-based synthetic data generation are considered to further balance the dataset.

2.3.4.2 Model Selection Experiments

The cast component adopts a **two-stage deep learning architecture** consisting of detection and classification phases. In the first stage, multiple object detection models including YOLOv8n, YOLOv8m, YOLOv10b, and RT-DETR are evaluated based on accuracy, precision, and computational complexity. Among these, **YOLOv8m** is selected due to its optimal balance between detection performance and efficiency. It is trained using manually annotated bounding boxes to accurately localize urinary casts within microscopy images. In the second stage, cropped cast regions are passed into classification models. Several CNN architectures, including **ResNet-18, MobileNet-V2, EfficientNet-B0, and EfficientNet-B2**, are evaluated for subtype classification. All models are initialized with pretrained ImageNet weights and fine-tuned on the cast dataset. Among these, **EfficientNet-B2** demonstrates the most consistent and reliable performance and is selected as the final classification model.

To improve classification performance and handle class imbalance, techniques such as **stratified five-fold cross-validation, inverse class weighting, and macro-F1 evaluation metrics** are applied. The final system integrates YOLOv8m for detection and EfficientNet-B2 for classification within a unified backend pipeline.

2.3.4.3 **RVO Component Model Architecture**

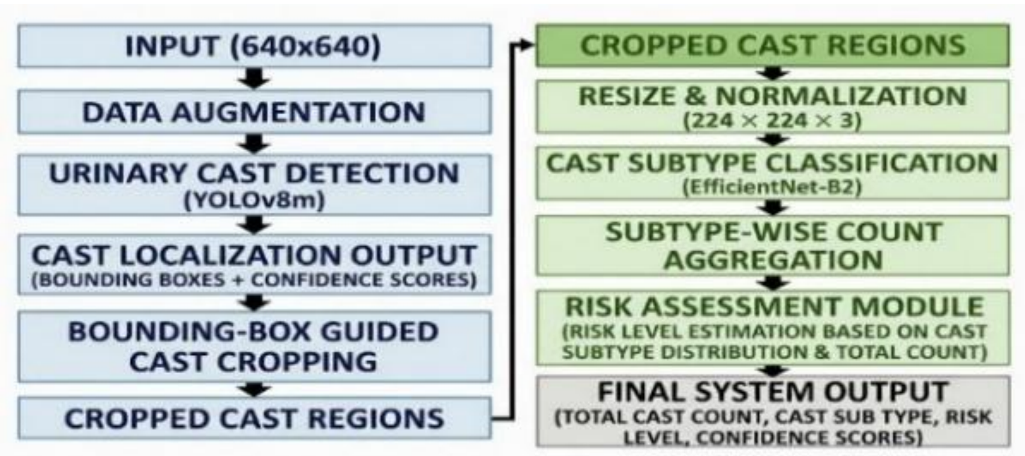

Figure 2.6: Architecture of the hybrid deep learning model for RVO classification

As shown in **Figure 2.6**, the architecture of the cast component follows a structured multi-stage pipeline. The process begins with input urine microscopy images, followed by preprocessing and data augmentation to improve model robustness. In the detection phase, the **YOLOv8m model** identifies and localizes cast regions using bounding box predictions, which are then used to crop relevant regions from the images. The cropped cast images are resized and normalized before being passed to the classification stage, where **EfficientNet-B2** is used to classify cast subtypes. The outputs are then aggregated to determine subtype distribution and total counts. Finally, a **risk assessment module** analyzes the detected cast types and their frequency to provide clinically meaningful insights related to kidney diseases.

This pipeline ensures accurate detection, reliable classification, and effective clinical interpretation, making the cast component a crucial part of the overall UroAI diagnostic framework.

2.4 Commercialization aspects of the product

The proposed **UroAI system** has strong commercialization potential as a scalable, AI-powered diagnostic support tool for urine microscopy analysis. The product is designed to address key limitations in traditional urinalysis, such as time consumption, dependency on skilled technicians, and variability in results. By automating the detection, classification, and clinical interpretation of urine sediment components, UroAI provides a cost-effective and efficient solution for healthcare institutions.

One of the primary commercialization opportunities lies in deploying UroAI as a **Software-as-a-Service (SaaS)** platform for hospitals, diagnostic laboratories, and clinics. Through a subscription-based model, healthcare providers can access the system via a web interface, enabling real-time analysis of urine microscopy images without requiring advanced hardware infrastructure. This makes the solution particularly valuable for **resource-limited settings and rural healthcare centers**, where access to expert analysis is limited. Additionally, the system can be integrated with existing **Laboratory Information Systems (LIS)** and hospital management systems to streamline workflows and improve diagnostic efficiency. The modular architecture of UroAI allows different components (UTI detection, kidney stone risk prediction, hematuria analysis, and cast-based renal disease diagnosis) to be offered as standalone services or as a complete diagnostic package, providing flexibility for different market needs.

From a market perspective, the growing demand for **AI-driven healthcare solutions** and early disease detection systems presents a significant opportunity. UroAI can be positioned as a **clinical decision support system**, assisting laboratory technicians and doctors rather than replacing them, thereby increasing acceptance in the medical field. Partnerships with hospitals, diagnostic labs, and healthcare providers can further enhance adoption and credibility. In terms of scalability, the system is designed to be deployed on **cloud platforms**, allowing efficient handling of large volumes of data and supporting multiple users simultaneously. Continuous model improvement can be achieved through feedback loops and additional data collection, enabling the product to evolve and maintain competitive performance.

Furthermore, regulatory compliance and data security are critical for commercialization. The system can be aligned with healthcare standards and data protection regulations to ensure safe handling of patient data. With proper validation and clinical testing, UroAI has the potential to be developed into a **certified medical diagnostic support tool**. Overall, UroAI presents a commercially viable solution with the potential to improve diagnostic accuracy, reduce workload, and enhance accessibility to quality healthcare services, making it suitable for large-scale adoption in modern healthcare systems.

Target Audience:

The target audience of the **UroAI system** includes:

- **Medical Laboratory Technicians (MLTs)** – to analyze urine microscopy images
- **Doctors (Nephrologists / General Physicians)** – to support diagnosis
- **Hospitals, clinics, and diagnostic labs** – to improve workflow and efficiency
- **Rural healthcare centers** – where expert analysis is limited
- **Patients** – to view reports and provide health information

Product Features:

The UroAI system provides automated analysis of urine microscopy images by detecting and classifying key sediment components such as RBCs, WBCs, bacteria, yeast, crystals, and casts. It supports accurate diagnosis of conditions including UTIs, kidney stone risk, hematuria causes, and renal diseases by combining image-based analysis with patient data such as symptoms and medical history. The system delivers real-time results with confidence scores and visual outputs like highlighted regions for better interpretability. Additionally, it is implemented as a user-friendly web-based platform, making it accessible, efficient, and suitable for practical clinical use.

2.5 Testing & Implementation

2.5.1 UTI Component Testing

The UTI component underwent extensive evaluation using multiple deep learning models for detection, segmentation, and classification tasks. The **YOLOv11-n model**, selected for WBC detection, achieved a **mean Average Precision (mAP@0.5) of 0.96**, demonstrating high accuracy in detecting infection-related particles. Yeast detection also showed strong performance with **mAP values exceeding 0.94**, while segmentation models such as **UNet++ (EfficientNet-B0 encoder)** and **SegFormer-B2** achieved high Dice and IoU scores, confirming accurate boundary segmentation.

The bacterial analysis module, implemented as a classification task using **EfficientNet-B0**, achieved **near-perfect validation accuracy**, indicating reliable differentiation between bacterial and non-bacterial samples despite the absence of bounding box annotations. In addition to image-based models, metadata-driven models were evaluated. While structured clinical data showed strong predictive capability, the symptom-based model demonstrated lower performance (**ROC-AUC ≈ 0.57**), highlighting the importance of microscopy-based evidence.

The integration of all outputs into a rule-based decision system was tested to classify UTI types accurately. The system successfully differentiated between bacterial and fungal infections as well as upper and lower UTIs. Implementation testing confirmed that the module performs efficiently in real-time, even when handling multiple image inputs simultaneously.

2.5.2 Kidney Stone Component Testing

The kidney stone component was evaluated using a two-stage pipeline consisting of crystal detection and classification. During testing, multiple detection models were compared, and **YOLO11m** demonstrated an optimal balance between detection accuracy, speed, and computational efficiency. The model achieved a **mAP@0.5 of approximately 0.91**, indicating strong performance in detecting small crystal structures within microscopy images.

In the classification stage, the **EfficientNet-B0 model achieved a test accuracy of 99.41% with a macro F1-score of 0.99**, demonstrating excellent performance in identifying crystal subtypes such as calcium oxalate and uric acid. Minor misclassifications were observed in cases where crystal structures had similar visual features. The integration of detection and classification results into a risk prediction module was also tested. The system successfully generated personalized kidney stone risk predictions based on crystal type and count. Implementation testing confirmed that the module operates efficiently with minimal latency, making it suitable for real-time clinical use.

2.5.3 Hematuria (RBC) Component Testing and Implementation

The hematuria component was tested using a two-stage approach involving RBC detection and morphological classification. The use of YOLO-based detection improved the quality of classification by isolating individual RBCs from microscopy images. Approximately **2,000 RBC samples** were used for evaluation, ensuring sufficient diversity in morphology. Among the evaluated models, **EfficientNet-B3 achieved the best performance**, with a **test accuracy of 81.40%, macro precision of 0.82, and recall of 0.81**. These results indicate improved sensitivity in detecting dysmorphic RBCs, which are clinically significant for identifying glomerular diseases.

Despite the improvements, challenges were observed due to visual similarities between isomorphic and dysmorphic RBCs and limitations in dataset size. However, the component demonstrated consistent performance and reliability for screening purposes. Implementation testing showed stable performance with real-time processing capability.

2.5.4 Cast Component Testing and Implementation

The cast component was evaluated using a two-stage pipeline combining object detection and subtype classification. The **YOLOv8m model achieved a mAP@0.5 score of 0.801**, demonstrating reliable detection of urinary casts in complex and noisy microscopy images. For classification, **EfficientNet-B2 achieved a validation accuracy of 74.48% and a macro F1-score of 0.6254**, indicating moderate but consistent performance across different cast subtypes. The system performed well for common cast types but showed lower accuracy for rare subtypes such as waxy casts due to class imbalance.

Implementation testing confirmed that the integrated pipeline successfully detects, counts, and classifies urinary casts, providing clinically meaningful insights for renal disease diagnosis. Data augmentation and cross-validation techniques improved model robustness, although further improvements are needed for rare classes.

2.5.5 UroAI Full System

Beyond individual components, the complete UroAI system was tested as an integrated platform to evaluate performance in real-world scenarios. The system was implemented using a **FastAPI backend**, enabling seamless communication between frontend interfaces and AI models. Testing was conducted in both **local environments and simulated cloud environments** to assess scalability and responsiveness.

Functional testing ensured correct data flow between modules, while performance testing evaluated system behavior under stress conditions such as **concurrent image uploads and high-resolution image processing**. The system demonstrated stable performance with acceptable response times, although minor delays were observed under heavy load conditions.

The modular architecture allowed independent execution of components, improving maintainability and scalability. This design ensures that individual models can be updated or replaced without affecting the overall system functionality.

2.5.5.1 **Model Services Implementation**

The model services in the proposed **UroAI system** are implemented as a centralized and modular backend pipeline that enables efficient execution of multiple AI models for urine microscopy analysis. The system is built using **FastAPI**, which acts as the core service layer responsible for handling incoming requests, managing model inference, and returning diagnostic results in real time. Each diagnostic component—UTI detection, kidney stone risk prediction, hematuria analysis, and cast classification—is deployed as an independent service within the backend, allowing scalable and flexible model integration.

When a user uploads a urine microscopy image, the backend service first performs preprocessing operations such as resizing, normalization, and format conversion. The processed image is then routed to the appropriate model services based on the requested analysis. For example, the UTI module invokes YOLO-based detection and segmentation models for WBCs and yeast, along with a classification model for bacteria. Similarly, the kidney stone module uses a two-stage pipeline where YOLO-based detection identifies crystals, and EfficientNet-based classification determines crystal types. The hematuria component uses YOLO detection followed by EfficientNet classification to analyze RBC morphology, while the cast component integrates YOLOv8m for detection and EfficientNet-B2 for subtype classification.

Each model service operates independently but follows a standardized input-output structure, producing structured results such as detected particle types, counts, bounding box coordinates, and classification labels. These outputs are then aggregated within the backend and passed to a decision-making layer, which applies clinically informed rules and generates final diagnostic predictions. This modular design ensures that models can be updated, replaced, or scaled without affecting other components of the system.

To ensure efficient performance, the model services are optimized for real-time inference using lightweight architectures and asynchronous request handling. The system supports concurrent processing of multiple requests, enabling smooth operation in clinical environments with high workloads. Additionally, logging and monitoring mechanisms are implemented to track model performance, detect errors, and support continuous improvement. Overall, the model services implementation provides a **robust, scalable, and efficient framework** for integrating multiple AI models into a unified diagnostic system, ensuring accurate and real-time analysis of urine microscopy data in practical healthcare settings.

2.5.5.2 **Backend Development and Integration**

The backend of the **UroAI system** is developed using **FastAPI (Python)** to handle image processing, model execution, and communication between system components. It manages API requests, routes data to relevant AI models, and returns prediction results in real time. All diagnostic modules (UTI, kidney stone, hematuria, and cast analysis) are integrated into a unified backend pipeline, enabling smooth data flow and efficient processing. The system supports modular integration, allowing each component to operate independently while contributing to the overall diagnosis. The backend is designed for scalability and performance, supporting concurrent requests and ensuring stable operation in real-world clinical environments.

2.5.5.3 **Frontend Implementation**

The frontend of the **UroAI system** is developed using **React.js** with **Tailwind CSS** to provide a responsive and user-friendly interface. It allows users to upload urine microscopy images, input patient details, and view diagnostic results in an organized manner. The interface supports different user roles such as medical laboratory technicians, doctors, and patients, ensuring easy interaction with the system. Visual outputs such as detected particles and highlighted regions are displayed to improve interpretability. The frontend communicates with the backend through API calls, enabling real-time data processing and seamless integration with the AI models.

2.5.5.4 **Testing Strategy**

The testing strategy of the **UroAI system** focuses on evaluating accuracy, performance, and reliability at both model and system levels. Each AI component was tested using separate training, validation, and test datasets to ensure unbiased evaluation. Model-level testing was conducted using metrics such as accuracy, precision, recall, F1-score, and mAP to assess detection and classification performance. System-level testing included functional testing to verify correct data flow, as well as performance testing under conditions such as multiple image uploads and high-resolution image processing. Additionally, integration testing was performed to ensure smooth interaction between frontend, backend, and AI modules. This comprehensive approach ensures that the system operates reliably and is suitable for real-world clinical use.

2.5.5.5 **Deployment and Cloud Testing**

The **UroAI system** is deployed using a scalable backend architecture, enabling real-time processing of urine microscopy images. The system was tested in both **local and cloud-based environments** to evaluate performance under real-world conditions. Cloud testing was conducted to assess scalability, availability, and system stability during multiple concurrent requests. The backend services were configured to handle high-resolution image uploads and parallel processing of AI models efficiently. The results confirmed that the system maintains stable performance with acceptable response times, making it suitable for deployment in clinical environments with varying workloads.

3. RESULTS AND DISCUSSIONS

3.1 Results

3.1.1 UTI Component Results and Discussion

3.1.2 Kidney Stone Component Results and Discussion

3.1.3 Hematuria (RBC) Component Results and Discussion

3.1.4 Cast Component Results and Discussion

3.2 Research Findings

3.2.1 UTI Component

3.2.2 Kidney Stone Component

3.2.3 Hematuria (RBC) Component

3.2.4 Cast Component

3.3 Results and Discussion of UroAI

The proposed **UroAI system** was evaluated across multiple diagnostic components, including UTI detection, kidney stone risk prediction, hematuria analysis, and urinary cast-based renal disease diagnosis. The results demonstrate that the system achieves strong performance across all tasks while maintaining computational efficiency suitable for real-time clinical applications. The modular architecture, which integrates detection, segmentation, and classification models, allows each component to operate effectively while contributing to a unified diagnostic framework.

In the UTI component, the system showed high accuracy in detecting infection-related particles. The YOLOv11-n model achieved a mAP@0.5 of 0.96 for WBC detection, while yeast detection exceeded 0.94, confirming reliable identification of key biomarkers. Segmentation models such as UNet++ and SegFormer-B2 provided accurate boundary extraction, and the bacterial classification model achieved near-perfect accuracy. However, symptom-based prediction showed lower reliability (ROC-AUC ≈ 0.57), highlighting the importance of microscopy-based analysis.

The kidney stone component demonstrated excellent performance using a two-stage architecture. The YOLO11m model achieved strong detection accuracy (mAP ≈ 0.91), while the EfficientNet-B0 classifier achieved a test accuracy of 99.41% and macro F1-score of 0.99. These results indicate highly reliable classification of crystal types, although minor confusion was observed in visually similar crystal structures.

For hematuria analysis, the EfficientNet-B3 model achieved a test accuracy of 81.40% with improved recall, indicating strong sensitivity in detecting dysmorphic RBCs. The use of YOLO-based detection improved classification accuracy by isolating individual RBCs. However, performance is still affected by dataset limitations and visual similarities between RBC types.

In the urinary cast component, the YOLOv8m model achieved a mAP@0.5 of 0.801 for detection, while the EfficientNet-B2 classifier achieved a validation accuracy of 74.48% and macro F1-score of 0.6254. The system performed well for common cast types but showed reduced accuracy for rare subtypes due to class imbalance and structural similarities.

Overall, the results confirm that the **UroAI system provides a reliable, efficient, and clinically interpretable solution for automated urine microscopy analysis**. Compared to traditional manual methods and isolated AI models, the proposed system offers improved diagnostic accuracy, faster processing, and better integration of multiple clinical factors. While some limitations remain, particularly in rare case detection and dataset diversity, the system demonstrates strong potential for real-world clinical deployment and future enhancement.

4. SUMMARY OF CONTRIBUTION

4.1 IT22221582

4.2 IT22589118

4.3 IT22595812

4.4 IT22590176

5. CONCLUSION

This research presents **UroAI**, an AI-powered system for automated urine microscopy analysis aimed at improving the diagnosis of urinary tract and renal diseases. The proposed system successfully integrates multiple deep learning models to detect, classify, and interpret key urine sediment components such as white blood cells, red blood cells, bacteria, yeast, crystals, and urinary casts. By combining particle-level analysis with clinically informed decision logic, the system provides a comprehensive and interpretable diagnostic framework.

The experimental results demonstrate that the system achieves strong performance across all components. The UTI module showed high accuracy in detecting infection-related particles, while the kidney stone component achieved near-perfect classification accuracy for crystal types. The hematuria component effectively differentiated RBC morphology, and the cast component provided reliable detection and classification of urinary casts. These results highlight the effectiveness of using a modular, multi-model approach for automated urinalysis.

In addition to accuracy, the system is designed for **real-time processing, scalability, and practical usability**, making it suitable for deployment in clinical environments. The integration of multiple diagnostic components into a single platform improves efficiency, reduces manual workload, and enhances consistency in results compared to traditional methods.

However, certain limitations were identified, including dataset size constraints, class imbalance in rare categories, and challenges in distinguishing visually similar structures. Addressing these limitations through larger and more diverse datasets, improved model optimization, and advanced data augmentation techniques will further enhance system performance.

In conclusion, the **UroAI system demonstrates strong potential as a reliable clinical decision support tool**, capable of transforming traditional urinalysis into a faster, more accurate, and scalable process. Future work will focus on expanding datasets, improving model generalization, and validating the system in real-world clinical settings to ensure its effectiveness and adoption in modern healthcare.

REFERENCES